Can a smaller giveaway make a better launch?

From social polarization to a product-launch decision

A hypothetical campus seller reaches 300 customers through word of mouth.
The aim: compare paid sales and contribution profit, not just publicity.

300
individual customers

36
tested policy choices

100
fresh confirmation markets

Mihir Jagdish Vasave (24B 4517) | Vaibhav Mathur (24B 4534)

Key result: fewer samples improved modeled profit; costly targeting was harder to justify.

MODEL_SPEC.md; confirmation_paired.csv

We kept the mechanism and made the question useful

1 INITIAL PITCH

Question: can changing contacts split opinions?

Keep: individual opinions, social contacts and disagreement.

2 FEEDBACK to DESIGN

Use a concrete application. Synthetic inputs are acceptable. Compare what-if scenarios.

Feedback summarized from our account.

3 FINAL QUESTION

For a bottle launch, which price and free-sample plan earns more?

Add: awareness, budget, perceived value and costs.

The application changed; the feedback between conversations and beliefs remained.

sources/project_pitch_social_polarization.pdf; Student-reported professor feedback; PROJECT_JOURNAL.md

The seller chooses the offer; each customer decides to buy

Seller decisions	Customer behavior	Measures
Price: Rs 400 / 500 / 600	Hear about launch; discuss appeal	Awareness: how many heard?
Free samples: 3 or 12	Buy once if value AND budget allow	Paid sales: samples excluded
Recipients: random / many contacts / across groups	Learn from owning the bottle	Profit after sold + free bottle costs

Boundary: 300 customers, 3 groups, 1,200 contacts, 60 abstract periods.
No stock limits, repeat purchases, competition or fixed overhead.

This is a decision experiment under stated assumptions, not a campus sales forecast.

MODEL_SPEC.md

Four simple AnyLogic parts create the agent-based model

Component in AnyLogic	What it does	What we can point to
Main agent	Creates market; schedules and records	Setup, counters, CSV export
Consumer population (300)	Stores each person’s budget, belief and ownership	One circle = one person
Cyclic Event: launchPeriod	10 attempts/tick; 300 attempts/period	Calls advanceLaunchPeriod()
Java rules + basic shapes	Discussion, buying, experience; draw results	Methods, circles, lines, text

Agent-based: total sales are the sum of individual decisions; no fitted sales-growth curve.

CampusProductLaunch_LiveNetwork.alp; anylogic/ProductKernel.java

We generated customer differences instead of collecting surveys

Input	Generation rule	Example mean	Example SD
Initial appeal	Beta(2,2); range 0–1	0.490	0.235
Value scale	Uniform Rs 500–1,000	Rs 770.12	Rs 144.19
Budget	Uniform Rs 400–1,000	Rs 710.93	Rs 178.23
Owner experience	Normal(0.8,0.1), clipped 0–1	0.803	0.094
Contacts/person	Generated graph; average 8	8.00	2.85

Example market: 322 / 1,200 ties cross groups (26.8%). SD = spread across its 300 customers.

Inputs are assumptions; repeated synthetic markets provide the output data.

data/customers_rep 1000.csv; input_statistics.csv; initial_graph_statistics.csv

Each behavior parameter answers a simple “how often / how much?”

Parameter	Baseline	Meaning / example
Sharing chance s	0.25	One aware person passes news in 1 in 4 eligible meetings
Tolerance d	0.20	Only belief gaps below 0.20 allow opinion averaging
Opinion adjustment μ	0.30	Move 30% toward the other’s old belief:0.40 to 0.43
Replacement chance φ	0.05	A disagreeing meeting has a 5% chance to replace its tie
New-contact rule h	0 or 1	0: random; 1: similar beliefs, random fallback
Experience q; unit cost	0.8; Rs 250	q is the normal draw’s pre-clipping mean; cost includes samples

A period is 300 interaction attempts—not a day. We do not claim these inputs are calibrated.

MODEL_SPEC.md; behavior_summary.csv

One conversation can inform; a purchase still needs two checks

SAMPLE	Free recipients start aware and use their own experience.
MEET	Select a person and one current contact; news may pass.
DISCUSS	Close beliefs move together; disagreement may replace the tie.
BUY / LEARN	At period end, eligible nonowners buy once and learn from use.

Example: Rs 800 × belief 0.60 = Rs 480 perceived value; budgetRs 450.
PriceRs 400 to buy. PriceRs 500 to do not buy.

Buy only if aware, not already an owner, budget ≥ price AND perceived value ≥ price.

MODEL_SPEC.md; anylogic/ProductKernel.java

We compare like with like, then check fresh markets

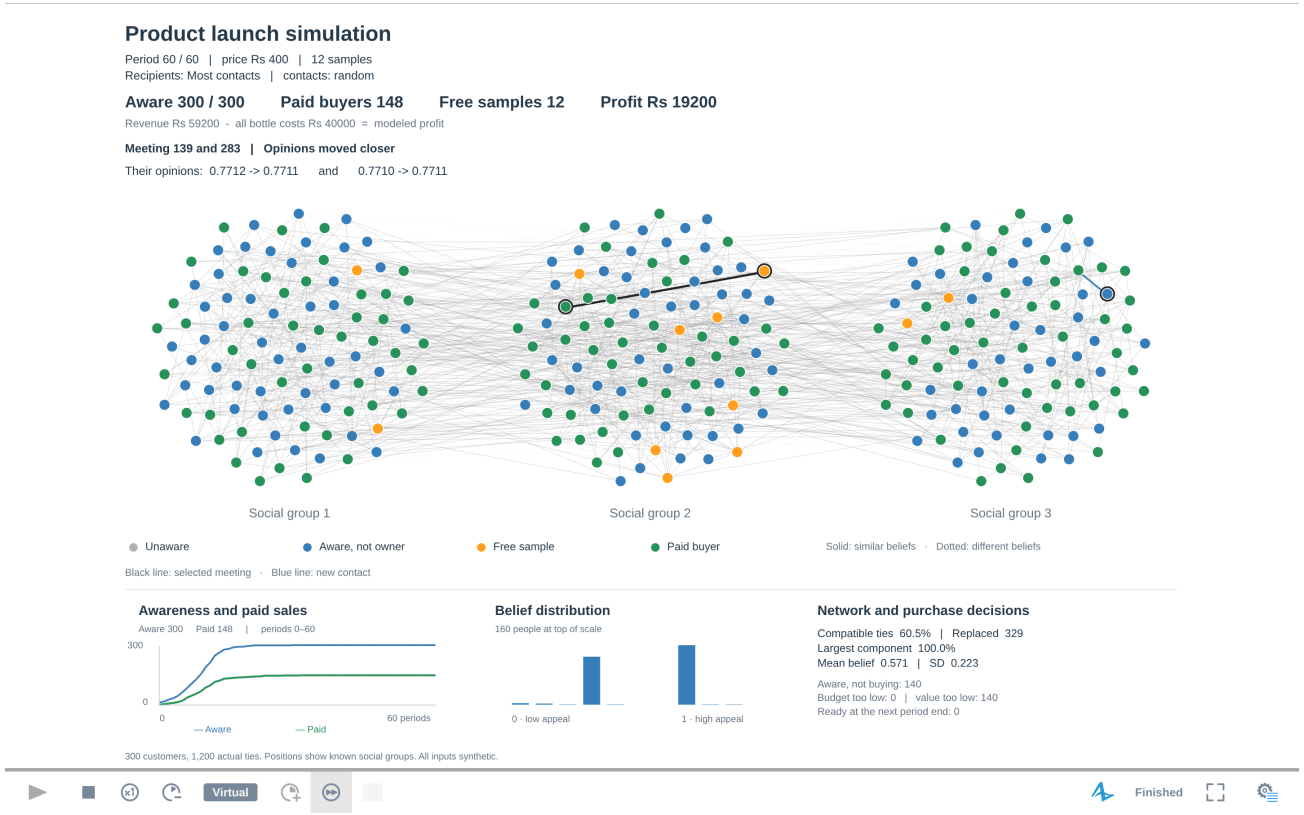
Stage	Markets × choices	Purpose
Pilot	30 separate markets ×18	Check setup and choose a provisional policy
Main + follow-up	100 markets ×36 =3,600 runs	Prices 3 × recipients 3 × contacts 2 × samples 2
Fresh confirmation	100 new markets ×4 =400 runs	Check fixed choices without searching again

Fair comparison: the same generated market is reused across policy alternatives.
95% interval: resample whole markets 10,000 times; never treat customers as replications.

The three-sample idea was exploratory. The separate fresh-market check strengthens its evidence.

sim/results/*manifest.json; confirmation_paired.csv

Live demo: follow news, one meeting and one purchase



Actual completed default run

Grey: unaware
Blue: aware, not owner
Orange: free sample
Green: paid buyer

Show one meeting and its belief change.
Then watch awareness and paid sales.

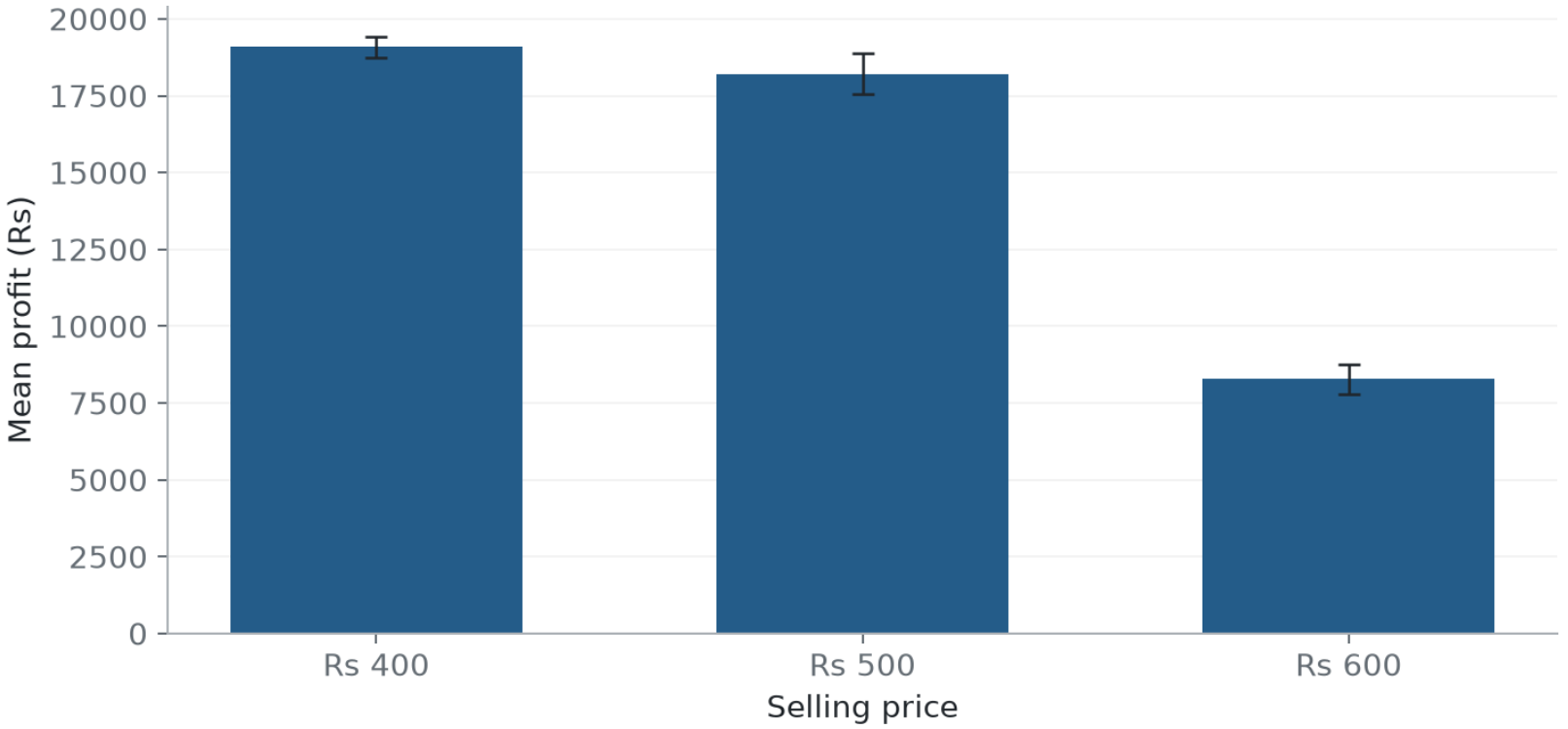
Policy 6 / rep 1000; price Rs 400
300 aware | 148 paid | 12 samples
Profit Rs 19,200
One run, not a policy average.

Solid / dotted ties explain agreement; node color explains customer state. These are different facts.

CampusProductLaunch_LiveNetwork.alp; anylogic/evidence/; LIVE_NETWORK.md

A higher price lost buyers faster than it gained margin

Price	Paid buyers	Mean profit
Rs 400	147.33	Rs 19,100
Rs 500	84.85	Rs 18,212
Rs 600	32.21	Rs 8,274



Near-total awareness is not launch success. Track paid sales and profit separately.

all_policy_summary.csv

Nine fewer free bottles raised profit by about Rs 2,846

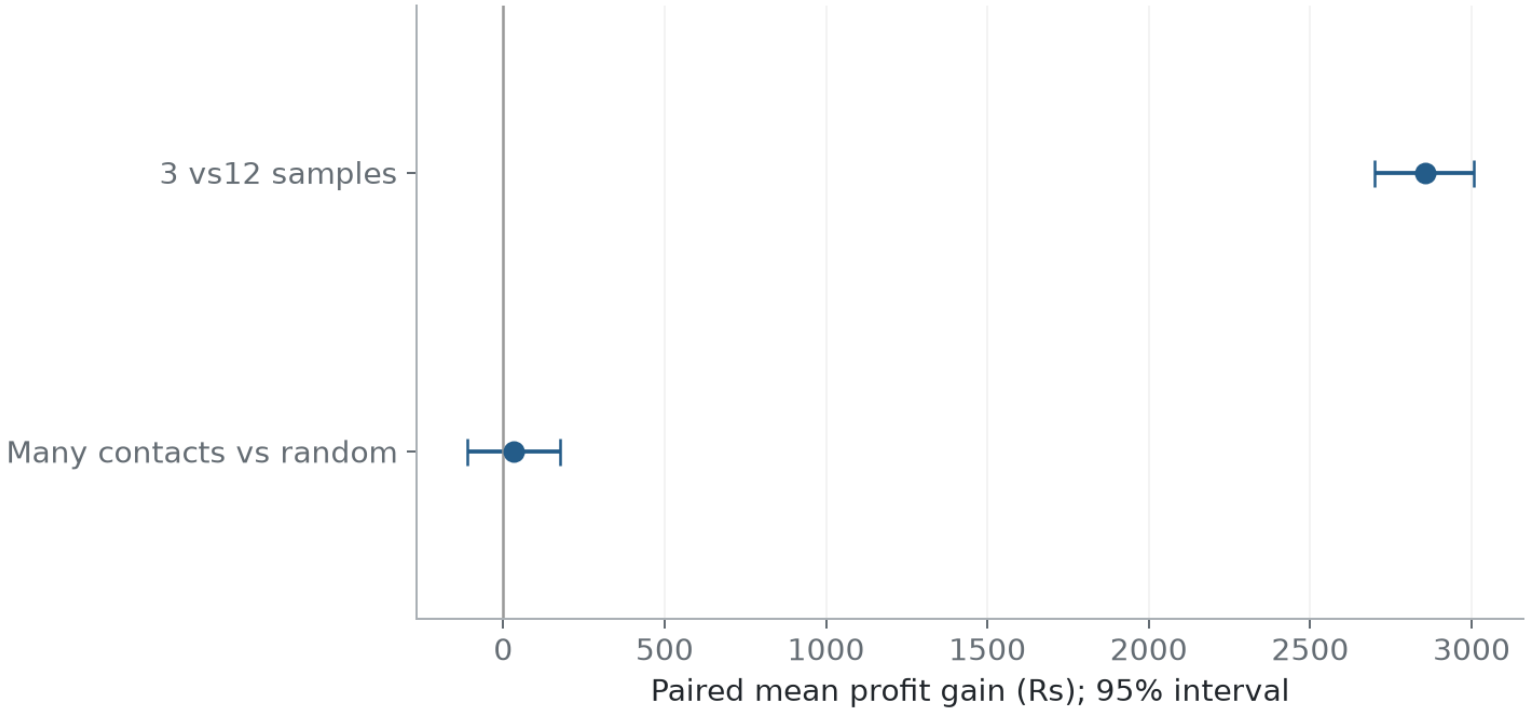
Outcome: 100 main markets	12 samples	3 samples
Paid buyers	147.33	151.30
Free bottles	12	3
Total owners (paid + free)	159.33	154.30
Mean profit	Rs 19,100	Rs 21,945

Profit gain = Rs 2,250 saved on samples + Rs 595.50 from extra paid margin.
Paired 95% interval: Rs 2,673 to Rs 3,013.50.

More owners and more profit are different objectives. Free bottles cost money.

budget_paired.csv; all_policy_summary.csv

Fresh markets support fewer samples—not expensive targeting



Rs 21,711

Mean profit: Rs 400, 3 samples, many-contact recipients

+Rs 2,854.50
Fewer samples: clear gain

+Rs 31.50
Targeting: uncertain gain

Do not pay a substantial targeting fee on the basis of this small, uncertain advantage.

confirmation_summary.csv; confirmation_paired.csv

What-if analysis changes the price and the timing advice

Unit cost	Best tested price	Mean profit
Rs 150	Rs 400	Rs 37,375
Rs 250	Rs 400	Rs 21,945
Rs 350	Rs 500	Rs 11,702
Rs 450	Rs 600	Rs 3,420

When groups are more separated

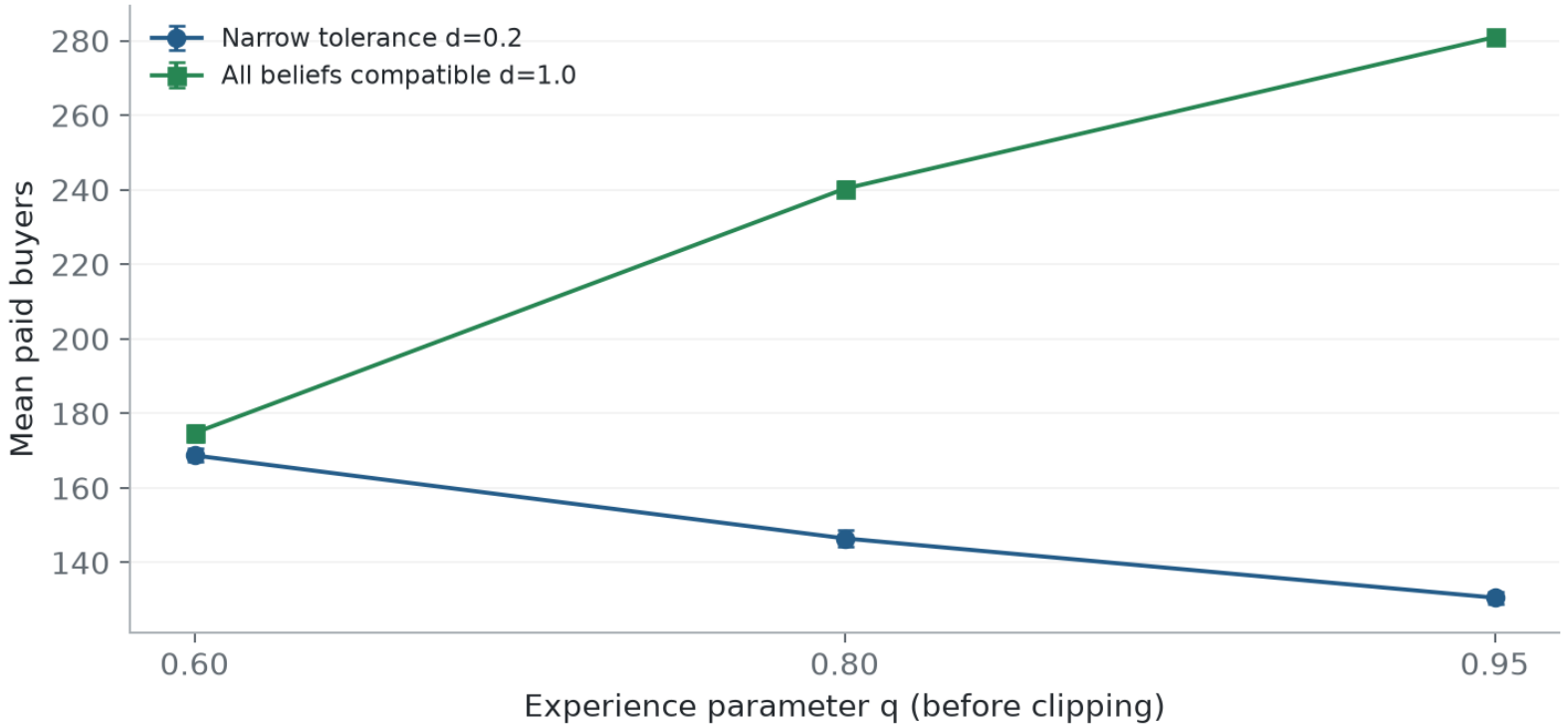
Across-group vs many-contact seeding:
Period 7: +Rs 393 [Rs 203, Rs 585]
Period 60: −Rs 53 [−Rs 155, Rs 50]

Similar-contact replacement atRs 400, 3 samples: −Rs 96 [−Rs 179, −Rs 15]. A small, exploratory effect; not a universal rule.

Reprice when costs change. Use broad group coverage when early reach is the objective.

all_cost_revaluation.csv; stress_paired.csv; final_paired_contrasts.csv

An awkward result exposed a weakness in our assumption



Why can this happen?

Belief 0.50 listens to 0.60.
It ignores 0.95 if d=0.20.

Very positive owners can become too far from moderate nonowners to influence them.

This is a model warning. Validate or soften the attention rule before real business use.

quality_diagnostic_summary.csv; quality_paired.csv

We improved the project by testing things that could go wrong

Issue	What we did	What we learned
Existing mechanisms looked “novel”	Read primary literature; credit prior models	Contribution is application + comparison
Free owners confused with sales	Separate free / paid flags and costs	Reach, ownership and profit differ
Unexpected quality reversal	1,200 diagnostic runs; tolerance / rewiring checks	Test assumptions; keep awkward results
Animation and run evidence differ	Preserve tested fallback; compare actual CSVs	A display/API check is not engine validation

Checks:16 product checks;1,080 standalone Java runs; actual matched default AnyLogic export.

PROJECT_JOURNAL.md; VALIDATION_REPORT.md; ../VALIDATION_REPORT.md

Start small; buy network information only if it pays

ACTION UNDER OUR ASSUMPTIONS

1 Test priceRs 400 and 3 free samples.

2 Use simple recipients unless contact information is cheap.

3 Revisit price when unit cost changes.

BEFORE APPLYING IT OUTSIDE CLASS

Validate budgets, sharing and buying rules.

Test the hard opinion cutoff.

Add targeting, inventory and operating costs.

No claim of real-campus demand.

Better decisions came from clear metrics, fair comparisons and honest limits—not a complex animation.

confirmation_paired.csv; all_cost_revaluation.csv; OPEN_ISSUES.md

Backup: exact parameters and policy codes

Setting	Definition / location
policyIndex =6	Rs 400;12 samples; many-contact recipients; random new contacts
policyIndex =24	Same as 6, but 3 samples; recommended-policy illustration
rep =1000	One reproducible Java market; not an average
Event recurrence =0.05	Animation scheduling time; not a market period/day
Model clock	10 attempts/visual tick;300 attempts/period;60 periods
Other defaults	N 300; edges 1200; d 0.2; μ 0.3; φ0.05; s 0.25; q 0.8

The simulation is a central event scheduler acting on individual agents; no physical movement model.

MODEL_SPEC.md; anylogic/ProductKernel.java; CampusProductLaunch_LiveNetwork.alp

Backup: statistical evidence and exact fresh-market comparisons

Provisional minus control	GainRs	95% paired intervalRs
3 vs 12 samples	2,854.50	2,698.50 to 3,006.00
Many contacts vs random	31.50	-112.50 to 177.00
vs Rs 500/12/random	4,108.50	3,581.50 to 4,623.50

100 independent fresh markets;4 fixed choices;400 evaluations.
Resampling unit = market. Confidence intervals do not validate real-world assumptions.

Observed outcome variation ≠ uncertainty about whether people actually behave this way.

confirmation_summary.csv; confirmation_paired.csv; sim/final_audit.json

Backup: published mechanisms and our own assumptions

Primary source	What it supports	What it does NOT validate
Deffuant et al.2000	Pairwise influence within a tolerance	Our purchase or budget rules
Kozma & Barrat 2008	Adaptive contacts and opinion clusters	A new discovery of fragmentation
Delre et al.2007 /2010	Product diffusion, awareness and seeding	Our exact monetary/customer parameters
Kempe et al.2003	Influence maximization; compare heuristics	An optimality guarantee for our profit rule

Read-status limits and claim corrections are explicit. Research, implementation and drafting were AI-assisted.

LITERATURE_NOTES.md; ../LITERATURE_NOTES.md; sources/project_pitch_social_polarization.pdf

Backup: where every evaluation criterion is covered

Professor's criterion	Slides / evidence
Scope and objectives	2–3: application, boundary and decisions
Data and input analysis	5–6: distributions, actual generated summaries and parameters
Model building	4,6–7: AnyLogic components, scheduling and worked individual rule
Basic + advanced output analysis	8,10–14: metrics, intervals, alternatives, cost and mechanism checks
Animation	9:90-second live agent-network demonstration
Recommendation + communication	15–16: lessons, practical advice and limits; sources in footer

Submit: .alp models + data/analysis support + slides. Keep the PDF and approved model as fallbacks.

SUBMISSION_GUIDE.md; COURSE_REQUIREMENTS.md